

# Quick start for LaTeXing with IEEEtran.cls for IEEE Computer Society Conferences

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**Abstract**—Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

## I. INTRODUCTION

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus enim. Vestibulum pellentesque felis eu massa. **TODO!**

The remainder of the paper starts with a presentation of related work (Section II). It is followed by a presentation of hints on L<sup>A</sup>T<sub>E</sub>X (Section III). Finally, a conclusion is drawn and outlook on future work is made (Section IV).

## II. RELATED WORK

Winery [1] is a graphical modeling tool. The whole idea of TOSCA is explained by Binz et al. [2].

To search for related work, ScienceOS<sup>1</sup> can be used; it stores data in the EU and backs its answers with citations, which you still have to read and check yourself. Writing hints for student theses are given in the open-access book *Student Thesis Projects*<sup>2</sup>. More hints are given in *Writing for Computer Science*<sup>3</sup>.

## III. L<sup>A</sup>T<sub>E</sub>X HINTS

This section contains hints on writing LaTeX. It focuses on minimal examples, which can be directly adapted to the content

### A. Handling of paragraphs

One sentence per line. This rule is important for the usage of version control systems. A new line is generated with a blank line. As you would do in Word: New paragraphs are generated by pressing enter. In LaTeX, this does not lead to a new paragraph as LaTeX joins subsequent lines.

<sup>1</sup><https://www.scienceos.ai/>

<sup>2</sup><https://vdf.ch/product/student-thesis-projects-en.html>

<sup>3</sup><https://doi.org/10.1007/978-1-4471-6639-9>

To start a new paragraph, press enter twice. This leads to an empty line. In word, there is the functionality to press shift and enter. This leads to a hard line break. The text starts at the beginning of a new line. In LaTeX, you can do that by using two backslashes (`\`). This is rarely used.

Please do *not* use two backslashes for new paragraphs. For instance, this sentence belongs to the same paragraph, whereas the last one started a new one. A long motivation for that is provided at <http://loopspace.mathforge.org/HowDidIDoThat/TeX/VCS/#section.3>.

#### Corresponding LaTeX code of ./paper.tex

```
644 %\begin{document}
645 One sentence per line.
646 This rule is important for the usage of version
    ↳ control systems.
647 A new line is generated with a blank line.
648 As you would do in Word:
649 New paragraphs are generated by pressing enter.
650 In LaTeX, this does not lead to a new paragraph
    ↳ as LaTeX joins subsequent lines.
651 To start a new paragraph, press enter twice.
652 This leads to an empty line.
653 In word, there is the functionality to press
    ↳ shift and enter.
654 This leads to a hard line break.
655 The text starts at the beginning of a new line.
656 In LaTeX, you can do that by using two
    ↳ backslashes (\textbackslash\textbackslash)
    ↳ .
657 \\
658 This is rarely used.
659
660 Please do \textit{not} use two backslashes for
    ↳ new paragraphs.
661 For instance, this sentence belongs to the same
    ↳ paragraph, whereas the last one started a
    ↳ new one.
662 A long motivation for that is provided at \url{
    ↳ http://loopspace.mathforge.org/
    ↳ HowDidIDoThat/TeX/VCS/#section.3}.
```

#### B. Notes separated from the text

The package `mindflow` enables writing down notes and annotations in a way so that they are separated from the main text.

---

This is a small note.

---

#### Corresponding LaTeX code of ./paper.tex

```
669 %\begin{document}
670 \begin{mindflow}
671 This is a small note.
672 \end{mindflow}
```

#### C. Handling TODOs

Marked text.

#### Corresponding LaTeX code of ./paper.tex

```
677 %\begin{document}
678 \textmarker{Marked text.}
```

With `\textmarker`, only the text color is changed, since this also works *well* for individual words.

Marked text.

#### Corresponding LaTeX code of ./paper.tex

```
683 %\begin{document}
684 \textcomment{Marked text.}{A comment on it.}
```

Manual marking for text that has been changed since the last version.

#### Corresponding LaTeX code of ./paper.tex

```
687 %\begin{document}
688 \modified{Manual marking for text that has been
    ↳ changed since the last version.}
```

This is some text. Changed text.

#### Corresponding LaTeX code of ./paper.tex

```
691 %\begin{document}
692 This is some text.
693 \change{FL1: text adjusted}{Changed text}.
```

Here is a comment.

#### Corresponding LaTeX code of ./paper.tex

```
696 %\begin{document}
697 Here is a comment\sidecomment{Comment}.
```

TODO!

#### Corresponding LaTeX code of ./paper.tex

```
700 %\begin{document}
701 \todo{A lot of text still needs to be produced
    ↳ here}
```

#### D. Hyphenation

LaTeX automatically hyphenates words. When using microtype, there should be fewer hyphenations than in other settings. You might still need to tweak the hyphenation. Here are some hints:

In case you write “application-specific”, then the word will only be hyphenated at the dash. You can also write `applica\allowbreak{ }tion-specific` (result: application-specific), but this is much more effort.

You can now write words containing hyphens which are hyphenated at other places in the word. For instance, `application"=specific` gets `application"=specific`. This is enabled by an additional configuration of the babel package.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

711 æƒüŧöŦ
712 In case you write \enquote{application-specific},
    ↳ then the word will only be hyphenated at
    ↳ the dash.
713 You can also write \verb!applica\allowbreak{}tion
    ↳ -specific1 (result: applica\allowbreak{}
    ↳ tion-specific), but this is much more
    ↳ effort.
714
715 You can now write words containing hyphens which
    ↳ are hyphenated at other places in the
    ↳ word.
716 For instance, \verb!application"=specific1 gets
    ↳ application"=specific.
717 This is enabled by an additional configuration of
    ↳ the babel package.

```

E. Typesetting Units

Numbers can be written plain text (such as 100), by using the siunitx package as follows: `100  $\frac{\text{km}}{\text{h}}$` , or by using plain L<sup>A</sup>T<sub>E</sub>X (and math mode): `100  $\frac{km}{h}$` .

Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

722 æƒüŧöŦ
723 Numbers can be written plain text (such as 100),
    ↳ by using the \href{https://ctan.org/pkg/
    ↳ siunitx}{siunitx} package as follows:
724 \SI{100}{\km\per\hour},
725 or by using plain \LaTeX{} (and math mode):
726 $100 \frac{\mathit{km}}{h}$ .

```

5% of 10 kg

Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

729 æƒüŧöŦ
730 \SI{5}{\percent} of \SI{10}{kg}

```

Numbers are automatically grouped: 123 456.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

733 æƒüŧöŦ
734 Numbers are automatically grouped: \num{123456}.

```

F. Surrounding Text by Quotes

Please use the “enquote command” to quote something. Quoting with “quote” or “quote” also works.

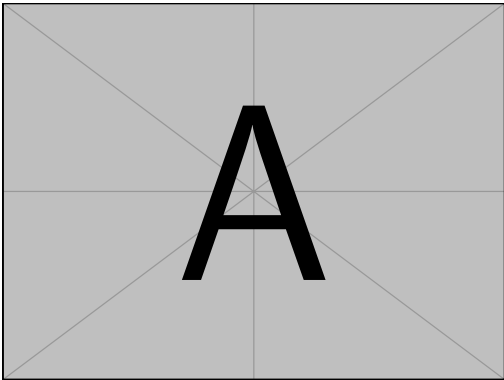


Figure 1. Example figure for zcref demo

| Heading1 | Heading2 |
|----------|----------|
| One      | Two      |
| Three    | Four     |

Figure 2. Example table for zcref demo

Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

739 æƒüŧöŦ
740 Please use the \enquote{enquote command} to quote
    ↳ something.
741 Quoting with "``quote'" or ``quote'" also works.

```

G. Zref-clever examples

zref-clever demonstration: Zcref (short for zcref[S]) at beginning of sentence, zcref in all other cases. Figure 1 shows a simple fact, although Figure 1 could also show something else. Figure 2 shows a simple fact, although Figure 2 could also show something else. Section III-G shows a simple fact, although Section III-G could also show something else.

Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

771 æƒüŧöŦ
772 \Zcref{fig:ex:cref} shows a simple fact, although
    ↳ \zcref{fig:ex:cref} could also show
    ↳ something else.
773
774 \Zcref{tab:ex:cref} shows a simple fact, although
    ↳ \zcref{tab:ex:cref} could also show
    ↳ something else.
775
776 \Zcref{sec:ex:cref} shows a simple fact, although
    ↳ \zcref{sec:ex:cref} could also show
    ↳ something else.

```

H. Figures

Figure 3 shows something interesting.

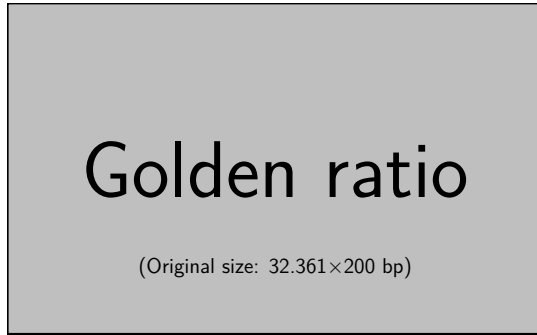


Figure 3. Simple Figure. Based on Scharrer [3].



Figure 4. Ulm Minster. Photo: wkrauth, CC0, via Wikimedia Commons.

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

781 %\begin{figure}
782 %\centering
783 %\includegraphics[width=.8\linewidth]{example-
784 %  image-golden}
785 %\caption[Simple Figure]{
786 %  Simple Figure.
787 %  Based on \cite{mwe}.
788 %}
789 %\label{fig:label}
790 %\end{figure}

```

Photos in JPEG or PNG format are included the same way, as shown in Figure 4.

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

797 %\begin{figure}
798 %\centering
799 %\includegraphics[width=.5\linewidth]{figures/
800 %  ulm-minster.jpg}
801 %\caption{Ulm Minster. Photo: wkrauth, CC0, via
802 %  Wikimedia Commons.}
803 %\label{fig:jpg}
804 %\end{figure}

```

One can span a figure across multiple columns by using `\begin{figure*}`. See Figure 5 as an example.

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

809 %\begin{figure*}
810 %\centering
811 % note that \textwidth is used instead of \
812 %  linewidth
813 % This ensures that the graphics width is 60%
814 %  of the "page" (text block), and not just
815 %  60% of the current text column
816 % See https://tex.stackexchange.com/a
817 %  /17085/9075 for details
818 %\includegraphics[width=.6\textwidth]{example-
819 %  image-16x9}
820 %\caption{16x9 Figure}
821 %\label{fig:16x9}
822 %\end{figure*}

```

#### I. Sub Figures

An example of two sub figures is shown in Figure 6.

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

826 %\begin{figure*}[!b]
827 %\centering
828 %\subfloat[Case I]{\includegraphics[width=.4\
829 %  linewidth]{example-image-a}}
830 %\label{fig:first_case}}
831 %\hfil
832 %\subfloat[Case II]{\includegraphics[width=.4\
833 %  linewidth]{example-image-b}}
834 %\label{fig:second_case}}
835 %\caption{Example figure with two sub figures.}
836 %\label{fig:two_sub_figures}
837 %\end{figure*}

```

Note that often IEEE papers with subfigures do not employ subfigure captions (using the optional argument to `\subfloat[]`), but instead will reference/describe all of them (a), (b), etc., within the main caption. Be aware that for `subfig.sty` to generate the (a), (b), etc., subfigure labels, the optional argument to `\subfloat` must be present. If a subcaption is not desired, leave its contents blank, e.g., `\subfloat[]`. An example is shown in Figure 7.

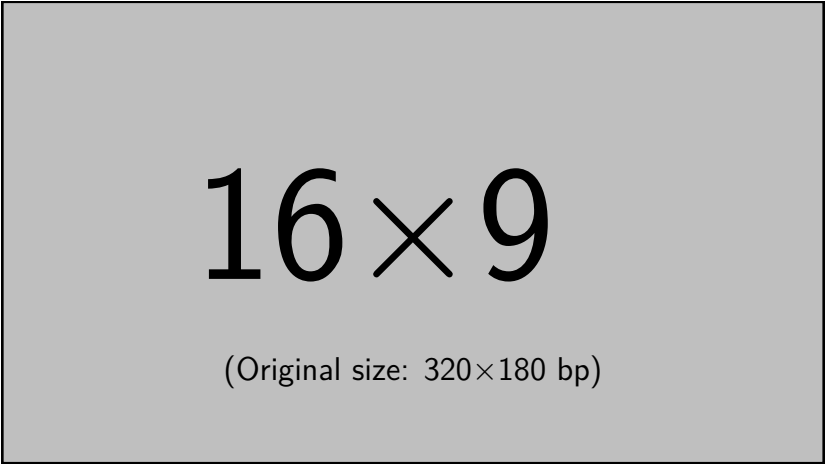
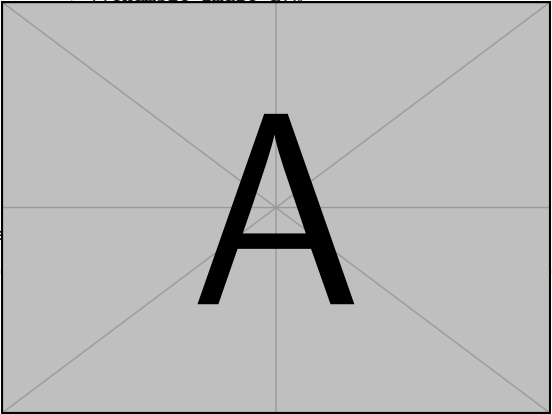


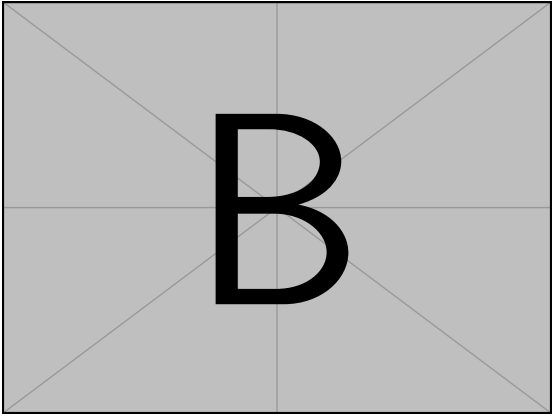
Figure 5. 16x9 Figure

Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```
848 æËÛöŦ
849 \begin{figure*}[!b]
850 \centering
851 \subfloat[]{\includegraphics[width=.4\linewidth
      ↳ ]{example-image-a}\%
852
853
854
855
856
857 \e
858
```

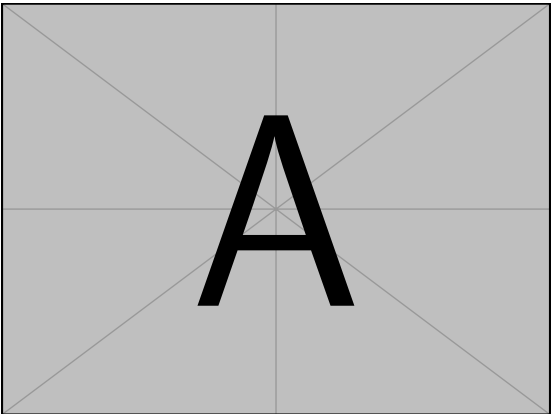


(a) Case I

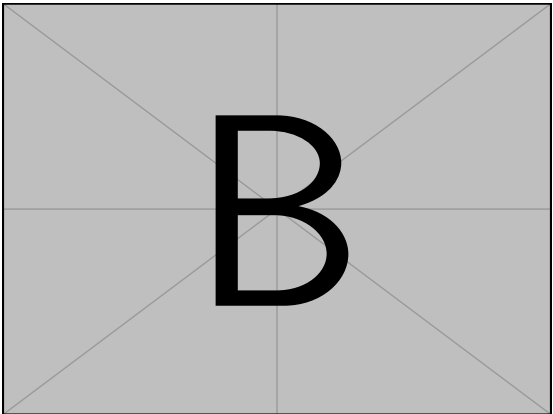


(b) Case II

Figure 6. Example figure with two sub figures.



(a)



(b)

Figure 7. Example figure with two sub figures. IEEE style. (a) The first case. (b) The second case.

### J. Figures with *tikz*

The *tikz* package creates graphics programmatically. This package generates grids and other regular structures.

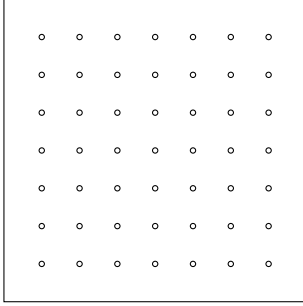


Figure 8. A regular grid generated easily with two for loops.

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

866 %\begin{figure}
867 \begin{figure}[ht]
868   \centering
869   \begin{tikzpicture}
870     \draw(0,0) rectangle (4,4);
871     \foreach \x in {0.5,1,1.5,2,2.5,3,3.5}
872       \foreach \y in {0.5,1,1.5,2,2.5,3,3.5}
873       \draw(\x,\y) circle (1pt);
874   \end{tikzpicture}
875   \caption{A regular grid generated easily with
876     \hookrightarrow two for loops.}\label{fig:tikz_example}
877 \end{figure}

```

### K. Plots with *pgfplots*

The *pgfplots* package plots functions directly in L<sup>A</sup>T<sub>E</sub>X, similar to MATLAB or gnuplot. Some visual examples are available in the package documentation<sup>4</sup>.

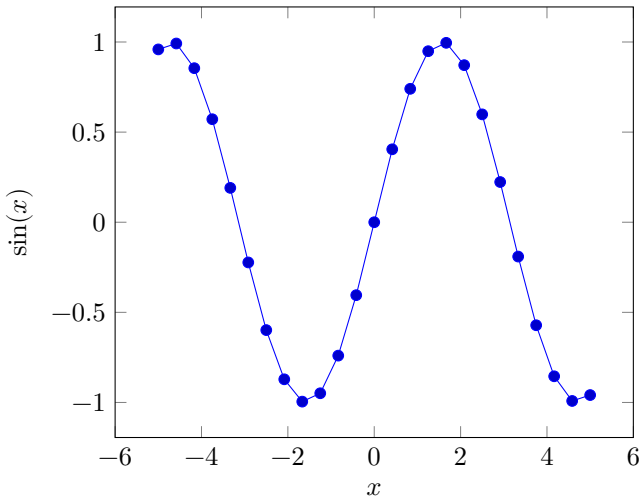


Figure 9. Plot of  $\sin(x)$  directly inside the figure environment with *pgfplots*.

<sup>4</sup><http://texdoc.net/pkg/pgfplots>

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

884 %\begin{figure}
885 \begin{figure}[ht]
886   \centering
887   \begin{tikzpicture}
888     \begin{axis}[xlabel=$x$, ylabel=$\sin(x)$]
889       \addplot {\sin(deg(x))}; % Plot the sine
890       \end{axis}
891     \end{tikzpicture}
892     \caption{Plot of $\sin(x)$ directly inside the
893       \hookrightarrow figure environment with pgfplots.}
894   \end{figure}

```

Coordinates can also be given explicitly instead of as a function:

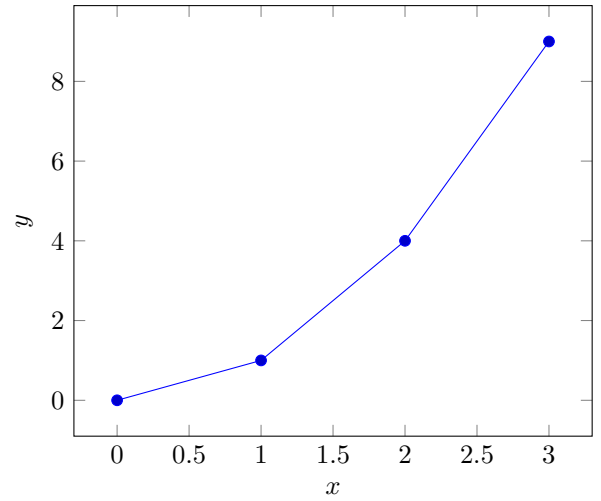


Figure 10. Explicit coordinates plotted with *pgfplots*.

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

898 %\begin{figure}
899 \begin{figure}[ht]
900   \centering
901   \begin{tikzpicture}
902     \begin{axis}[xlabel=$x$, ylabel=$y$]
903       \addplot coordinates {(0,0) (1,1) (2,4) (3,
904         \hookrightarrow 9)}; % Plot explicit coordinates
905     \end{axis}
906   \end{tikzpicture}
907   \caption{Explicit coordinates plotted with
908     \hookrightarrow pgfplots.}
909   \label{fig:pgfplots_coors}
910 \end{figure}

```

### L. Tables

Note that IEEE does not support `\begin{table}`, one has to use `\begin{figure}`.

Figure 11. Simple Table

| Heading1 | Heading2 |
|----------|----------|
| One      | Two      |
| Three    | Four     |

Figure 12. Table with diagonal line

|                       |                        |        |       |
|-----------------------|------------------------|--------|-------|
| Diag<br>Column Head I | Diag Column<br>Head II | Second | Third |
|                       |                        | foo    | bar   |

### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

915 %\begin{figure}
916 \begin{figure}
917   \caption{Simple Table}
918   \label{tab:simple}
919   \centering
920   \begin{tabular}{ll}
921     \toprule
922     Heading1 & Heading2 \\
923     \midrule
924     One      & Two      \\
925     Three   & Four     \\
926     \bottomrule
927   \end{tabular}
928 \end{figure}

```

### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

931 %\begin{figure}
932 % Source: https://tex.stackexchange.com/a
933 % ↪ /468994/9075
934 \begin{figure}
935   \caption{Table with diagonal line}
936   \label{tab:diag}
937   \begin{center}
938     \begin{tabular}{|l|c|c|}
939       \hline
940       \diagbox[width=10em]{Diag \Column Head I}{
941         ↪ Diag Column\Head II} & Second &
942         ↪ Third \\
943       \hline
944     \end{tabular}
945   \end{center}
946 \end{figure}

```

```

1 <listing name="example">
2   Floating
3 </listing>

```

Listing 2. Example XML listing – placed as floating figure

```

1 {
2   key: "value"
3 }

```

Listing 3. Example JSON listing – placed as floating figure

## M. Source Code

Listing 1 shows source code written in XML. Line 2 contains a comment.

```

1 <listing name="example">
2   <!-- comment -->
3   <content>not interesting</content>
4 </listing>

```

Listing 1. Example XML Listing

### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

951 %\begin{figure}
952 \Zcref{lst:XML} shows source code written in XML.
953 \Zcref{line:comment} contains a comment.
954
955 \begin{lstlisting}[
956   language=XML,
957   caption={Example XML Listing},
958   label={lst:XML}]
959 <listing name="example">
960   <!-- comment --> (* \label{line:comment} *)
961   <content>not interesting</content>
962 </listing>
963 \end{lstlisting}

```

One can also add float as parameter to have the listing floating. Listing 2 shows the floating listing.

### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

969 %\begin{figure}
970 \begin{lstlisting}[
971   % one can adjust spacing here if required
972   % aboveskip=2.5\baselineskip,
973   % belowskip=-.8\baselineskip,
974   float,
975   language=XML,
976   caption={Example XML listing -- placed as
977     ↪ floating figure},
977   label={lst:flXML}]
978 <listing name="example">
979   Floating
980 </listing>
981 \end{lstlisting}

```

One can also typeset JSON as shown in Listing 3.

```

1 public class Hello {
2     public static void main (String[] args) {
3         System.out.println("Hello World!");
4     }
5 }

```

Listing 4. Example Java listing

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

986 %\begin{lstlisting}
987 \begin{lstlisting}[
988     float,
989     language=json,
990     caption={Example JSON listing -- placed as
991             ↳ floating figure},
991     label={lst:json}]
992 {
993     key: "value"
994 }
995 \end{lstlisting}

```

Java is also possible as shown in Listing 4.

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

1000 %\begin{lstlisting}
1001 \begin{lstlisting}[
1002     caption={Example Java listing},
1003     label=lst:java,
1004     language=Java,
1005     float]
1006 public class Hello {
1007     public static void main (String[] args) {
1008         System.out.println("Hello World!");
1009     }
1010 }
1011 \end{lstlisting}

```

### N. Itemization

One can list items as follows:

- Item One
- Item Two

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

1018 %\begin{itemize}
1019 \begin{itemize}
1020     \item Item One
1021     \item Item Two
1022 \end{itemize}

```

With the package paralist, one can create itemizations with lesser spacing:

- Item One
- Item Two

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

1027 %\begin{compactitem}
1028 \begin{compactitem}
1029     \item Item One
1030     \item Item Two
1031 \end{compactitem}

```

One can enumerate items as follows:

- 1) Item One
- 2) Item Two

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

1036 %\begin{enumerate}
1037 \begin{enumerate}
1038     \item Item One
1039     \item Item Two
1040 \end{enumerate}

```

With the package paralist, one can create enumerations with lesser spacing:

- 1) Item One
- 2) Item Two

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

1045 %\begin{compactenum}
1046 \begin{compactenum}
1047     \item Item One
1048     \item Item Two
1049 \end{compactenum}

```

With paralist, one can even have all items typeset after each other and have them clean in the TeX document:

1) All these items... 2) ...appear in one line 3) This is enabled by the paralist package.

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

1054 %\begin{inparaenum}
1055 \begin{inparaenum}
1056     \item All these items...
1057     \item ...appear in one line
1058     \item This is enabled by the paralist package.
1059 \end{inparaenum}

```

### O. Other Features

The words “workflow” and “dwarflike” can be copied from the PDF and pasted to a text file.

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```

1064 %\enquote{workflow} and \enquote{
1065 The words \enquote{workflow} and \enquote{
1066     ↳ dwarflike} can be copied from the PDF and
1067     ↳ pasted to a text file.

```

The symbol for powerset is now correct:  $\wp$  and not a Weierstrass  $p$  ( $\wp$ ).

$\wp(1, 2, 3)$

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```
1068 æẼũŰőŰ
1069 The symbol for powerset is now correct: $\
      ↪ powerset$ and not a Weierstrass p ($\wp$)
      ↪ .
1070
1071 $\powerset(\{1,2,3\})$
```

Brackets work as designed: <test> One can also input backticks in verbatim text: `test`.

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```
1074 æẼũŰőŰ
1075 Brackets work as designed:
1076 <test>
1077 One can also input backticks in verbatim text: \
      ↪ verb|`test`|.
```

Acronyms can be typeset in running text with `\initialism`, which sets them as small caps that blend into the surrounding lowercase text. It is the lightweight alternative to the full acronym and glossary management (`\gls`, with an automatic list of abbreviations) provided by the thesis variants: `\initialism` only typesets the acronym and does not add it to any list. The macro and a few ready-made acronyms (`\OMG`, `\BPEL`, `\BPMN`, `\UML`) are defined in `commands.tex`, where you can add your own.

The UML is maintained by the OMG; related standards include BPEL and BPMN. Any acronym can be set inline with REST or HTTP.

#### Corresponding L<sup>A</sup>T<sub>E</sub>X code of ./paper.tex

```
1082 æẼũŰőŰ
1083 The \UML is maintained by the \OMG; related
      ↪ standards include \BPEL and \BPMN.
1084 Any acronym can be set inline with \initialism{
      ↪ REST} or \initialism{HTTP}.
```

## IV. CONCLUSION AND OUTLOOK

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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## ACKNOWLEDGMENT

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which is placed just before the reference section in your document [4].

## REFERENCES

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All links were last followed on October 5, 2020.